

Air Quality Effects on Health and Mortality rate

Group 4 members:

Yiheng Sun

Golnaz Berenjian

Navdeep Kaur Grewal

Amina Mahfoud

Introduction of Air Pollution:

Air pollution, driven by sources like industrial emissions and vehicle exhaust, poses significant health risks, contributing to various diseases. Our research focuses on seven key pollutants (**PM2.5, PM10, NO2, O3, SO2, CO, and NH3.**) that are central to the issue. While air pollution also affects climate change, this study specifically examines its impact on human health.

Research questions:

- Are there specific pollutants (e.g., PM2.5) that show a stronger correlation with adverse health outcomes?
- Which countries have the most impact of health issues throughout the world?
- Is there a big difference between the most polluted country vs the least polluted country?
- Are there any other factors that might explain the discrepancy between air pollution levels and mortality rates in countries?
- Are there significant differences in mortality rates between different countries?

API:

- **Countries & Pollutants:** OpenWeatherMap
- **Mortality Rate:** World Health Organization (WHO)
- **Hygiene level:** World Bank

Since the air pollution data was based on cities and the mortality rate data was based on countries, an important step in data exploration was aligning these datasets. By matching the common countries of the cities provided by OpenWeatherMap and the countries of mortality Rate by WHO.

By converting the format of country codes to ensure the two-dataset merge perfectly.

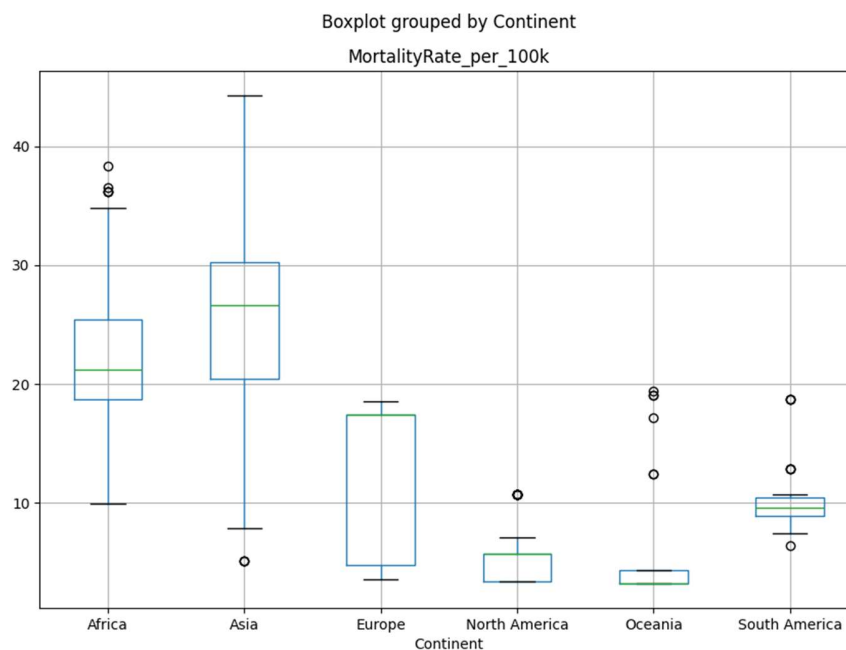
Hypothesis testing:

Question: Are there significant differences in mortality rates between different countries / continents?

- **Null Hypothesis:** There is no significant difference in mortality rates between countries /continents vs. Mortality rates.
- **Alternative Hypothesis:** There is a significant difference in mortality rates between countries/continents vs. Mortality rates.

Solution:

The countries dataset is split into 6 groups (continents) and a boxplot is generated to compare the means. **Anova One-way** test is deployed to retrieve the P-Value.



F_onewayResult: (statistic=121.45747674065453), (P-value=2.1944559322042223e-77)

Interpretation:

The p-value is extremely small (2.19×10^{-77}), which is effectively 0. A p-value this small strongly indicates that **the null hypothesis can be rejected**. This means that there is significant evidence that the mortality rates differ across the countries, depending on the continent that each country is located in.

Analysis:

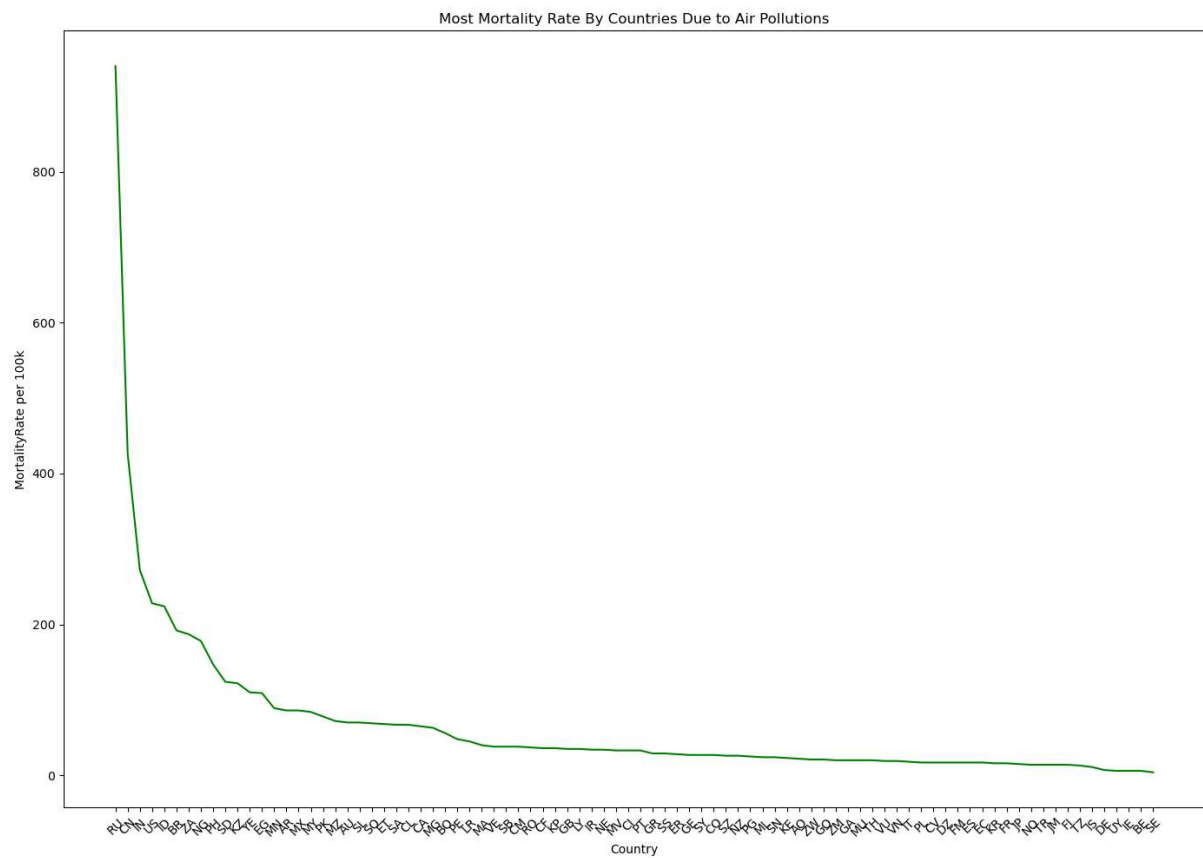


Fig 1. Total country ranking by Mortality Rate

By converting merged the data frame from csv file into a ranking data frame, with highest mortality rates to lowest mortality rates, sorted by country.

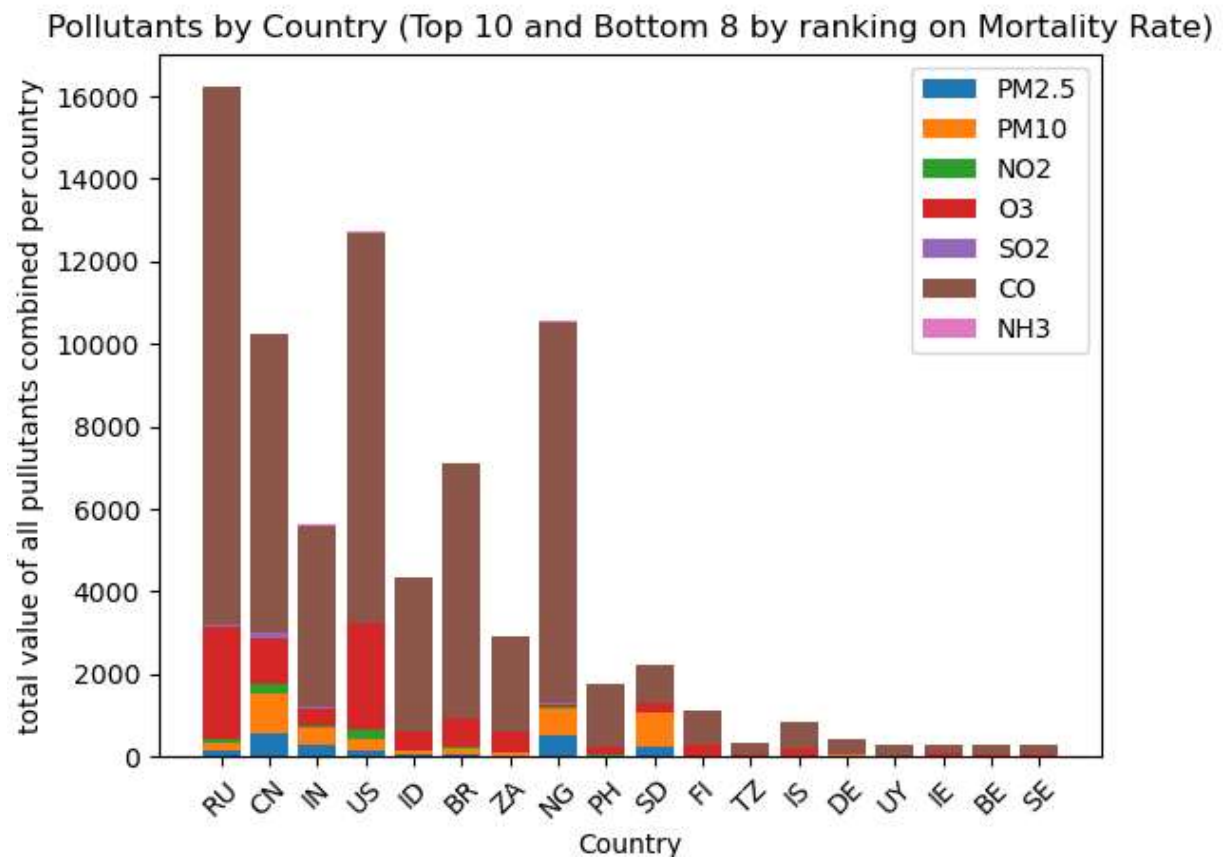


Fig 2. Comparison between top 10 and bottom 10 mortality rate countries with sum of all air pollutants by each country

Linear Regressions of each pollutants

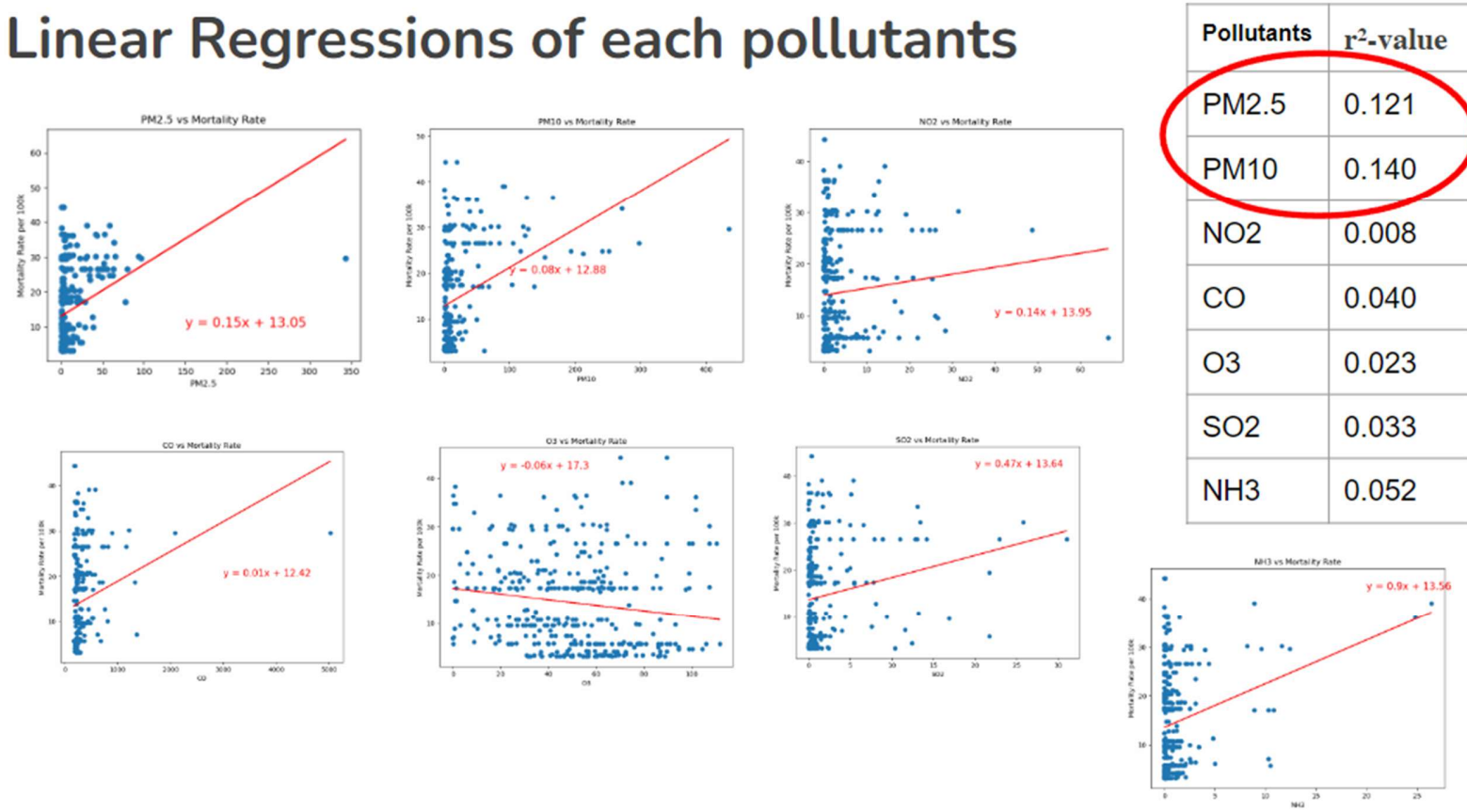


Fig 3. Seven different pollutants in linear Regression with tables of r²-value (left top: PM2.5, mid top: PM 10, right top: NO2, bottom left: CO, bottom mid: O3, bottom mid second: SO2, bottom right: NH3)

As the linear Regression shows in seven pictures inside, most of the pollutants show a positive increase on the slope by meaning of the more air pollutants values, the more mortality rate it could have. However, O3 with a slight drop as expected but can be explained into O3 is not a strong toxic pollutant.

The table of r²-value shows each pollutant corresponding to the mortality rate, the higher the r²-value, the more corresponding to the mortality rate. In fig 2. CO is the most air pollutant over all the countries, however, the PM2.5 and PM10 with higher r²-value proves more corresponding to the mortality rate, Moreover, those two pollutants is more toxic than other two air pollutants which make the results solid.

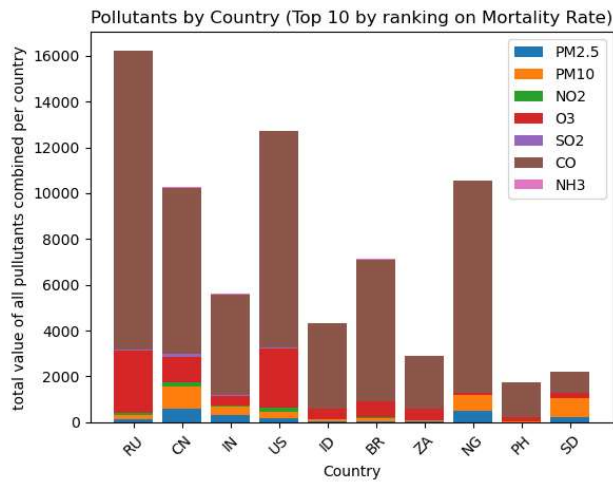


Fig 4. Top 10 countries ranked by mortality rate due to air pollutions with sum of pollutants

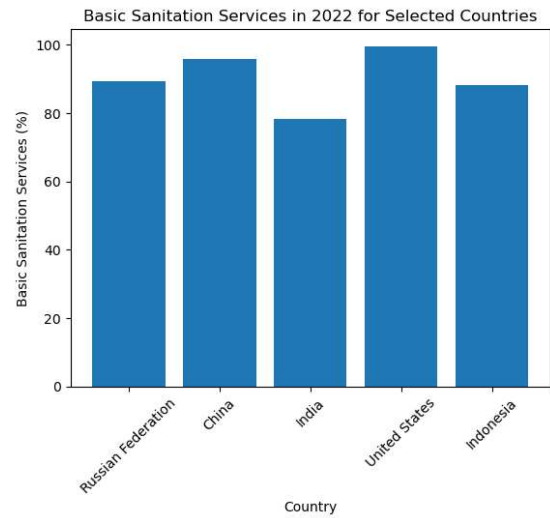


Fig 5. Basic Sanitation services corresponding to Fig 4. Top 5 countries

During the analysis (Fig 4), India has more mortality rates than United States, but with lower air pollutants values in total. The other resources from WORLD BANK with hygiene in different countries may help to understand and improve the analysis. In Fig 5, India has the lowest basic sanitation services according to the top 5 mortality rate countries.

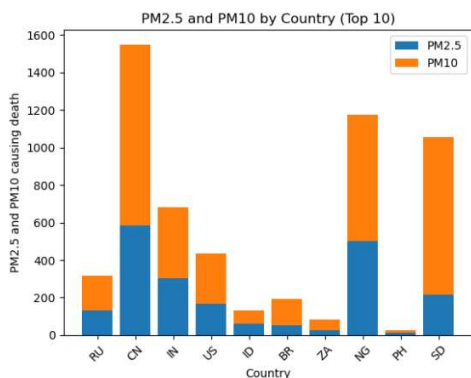


Fig 6. Top 10 mortality countries sorted by PM2.5 and PM10 pollutants

Moreover, PM2.5 and PM10 is more toxic pollutants, with higher r^2 -value that analyzed from linear regression. Also prove the reason why India has less total air pollutions but have higher mortality.

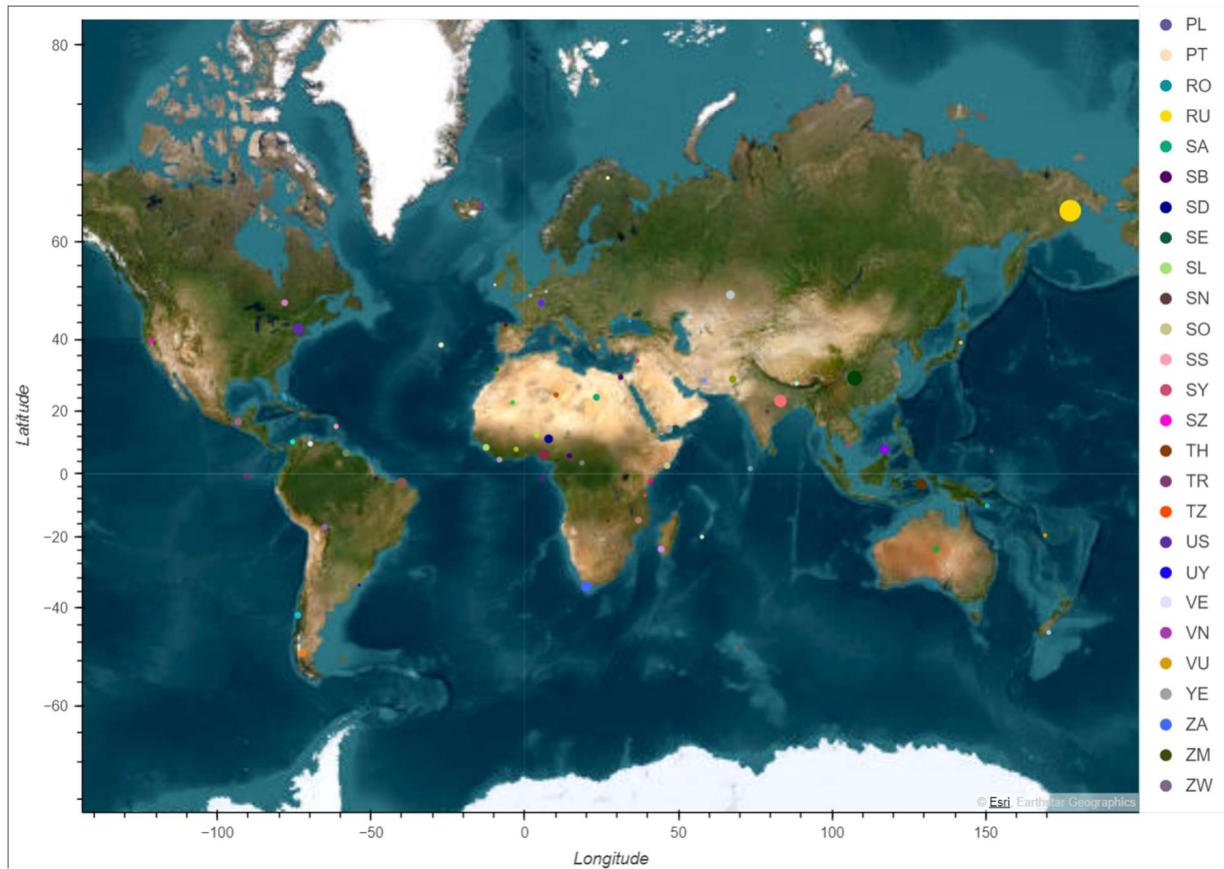


Fig 7. Geo Graph shows the mortality rate sorted by countries.

Conclusion:

In conclusion, although CO may have the highest pollutant levels, PM10 and PM2.5 show the strongest correlations with increased mortality rates. Additionally, other factors, such as hygiene, also play a significant role in influencing mortality, complicating the relationship between pollution levels and health outcomes. These findings suggest that while air pollution is a key factor, broader environmental and public health conditions must be considered when addressing mortality risks.